

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Engineering Economics (New)

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Discuss Engineering Economics and describe Principles and application of Engineering Economics. 7
- b) Define with a suitable example about the opportunity cost, marginal cost and sunk cost? Also explain manufacturing and non-manufacturing cost. 8
2. a) Ghanshyam is planning for his retirement. He is going to place a deposit Rs. 30000 per month in Citizen Investment Trust (CIT). CIT will give an interest rate of 7.5% compounded quarterly. If he continues his deposit for 15 years from now how much money he will have in his account? 7

OR

Mr. Gurung needs Rs. 1500000 at the end of 15th year for the study of his daughter. How much money does he need to deposit every year in a bank account for 10 years if interest rate is 9% compounded annually?

- b) Calculate both types of BCR using AW and FW formulation for the following Project. 8

Initial Investment	500000
Annual Revenue	10000
Annual Cost	15000
Useful life	8 years
Salvage Value	50000
MARR	12%

3. a) Compute IRR and suggest the investment decision and also draw unrecovered investment diagram for the project whose cash-flow given below: 8

Initial investment: Rs 90000
Annual Revenue: Rs 27000
Annual Expenses: Rs 8000
Salvage Value: Rs 10000
Expected Life: 6 year
MARR: 15%

- b) Evaluate whether the project is feasible for investment or not using Payback period method? Find both types of Payback period. 7
- Initial investment: Rs. 400000
 Annual benefit: Rs. 75000
 Annual cost: Rs. 25000
 Salvage value: Rs. 100000
 Useful life: 10 years
 Interest rate: 10% per year
4. a) Perform Sensitivity analysis of the following project over a range of $\pm 30\%$ in a) Initial Investment b) Net annual income and c) Useful life using PW formulation. MARR = 10%. Draw Sensitivity diagram also. 8

Initial Investment	20000
Annual Revenue	5000
Annual Cost	1000
Useful life	8 years
Salvage Value	2000

- b) From the following information, find economic service life of an assets. 7
- Initial Cost: Rs. 50,000
 Annual Operating cost: Rs 10,000 for the first year and increases by 15% each year
 Salvage value: Declining each year by 20% from the previous year's salvage value
 Useful life: 7 yrs
 MARR: 15%

OR

Explain the reasons for replacement of assets. The Annual Equivalent Cost (AEC) of the defender and challenger are given in the table below. What is the best replacement strategy? Use MARR: 12% per year and the planning horizon of the project is 8 years.

End of Year (n)	1	2	3	4	5	6
(AEC) _D (Rs)	5,300	5,250	5,400	5,750	6,200	6,550
(AEC) _C (Rs)	7,700	6,150	5,700	5,600	5,675	5,800

5. a) Compute the annual depreciation allowance and resulting book value using 8
- SOYD method
 - Sinking fund method if the Initial cost of equipment is Rs 10,000, Salvage value is Rs 2,000 and useful life is 5 years. Assume MARR = 12%.
- b) What do you understand by capital budgeting? Explain the different methods of financing an engineering project with its advantages and disadvantages. 7

6. a) What are the reasons of inflation? Explain actual dollar and constant dollar. 7
- b) Evaluate the following two feasible investment A and B having different useful lives, if MARR is 10% per year. Use PW method with repeatability assumption. 8

<u>Project: A</u>	<u>Project: B</u>
Initial investment: Rs 50000	Initial investment: Rs 150000
Annual Revenue: Rs 25000	Annual Revenue: Rs 70000
Annual Expenses: Rs 3000	Annual Expenses: Rs 2000
Salvage Value: Rs 15000	Salvage Value: Rs 40000
Expected Life: 3 year	Expected Life: 5 year

7. Write short notes on: (**Any two**) 2×5
- a) Causes of project risk
- b) Value added tax
- c) Time value of money